

Seoyul Oh

Ph.D. Candidate at the University of Illinois Urbana-Champaign (UIUC)

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Interests

My research focuses on advancing the efficiency, reliability, and safety of networked systems through the application of game-theoretic models and practical machine learning techniques.

Education

University of Illinois Urbana-Champaign, IL, USA

Sep 2023 - Current

Doctor of Philosophy, Computer Science

Advisors: [Prof. Francis Y. Yan](#) & [Prof. Deepak Vasisht](#)

Korea University, Seoul, Korea

Feb 2023

Master of Science, Electrical and Computer Engineering

Advisor: [Prof. Sangheon Pack](#)

Korea University, Seoul, Korea

Feb 2021

Bachelor of Science, Electrical Engineering

Publications

- [1] **A Call for Decentralized Satellite Networks** 📄
[Seoyul Oh](#), Deepak Vasisht
ACM Workshop on Hot Topics in Networks HotNets 2024
- [2] **EcoCell: Energy-aware Traffic Shaping for Cellular Radio Access Networks** 📄
Zikun Liu, [Seoyul Oh](#), Bill Tao, Anuj Kalia, Yaxiong Xie, Deepak Vasisht
New Ideas in Networked Systems NINeS 2026
- [3] **A First-Principles Diagnosis of RL Training in Networked Systems** 📄
[Seoyul Oh](#)^{*}, Yikai Zheng^{*}, Deepak Vasisht, Francis Y. Yan
Under Review, ^{*}Equal contribution Under Review
- [4] **OrbitShare: A Multi-party Architecture for Decentralized LEO Satellite Networks** 📄
[Seoyul Oh](#), Bill Tao, Shreya Kannan, Deepak Vasisht
Under Review Under Review
- [5] **An Efficient Winner and Payment Determination Algorithm in Reverse Auction for Edge Federation** 📄
Taeyun Kim, Joonwoo Kim, Hochan Lee, [Seoyul Oh](#), Sangheon Pack
IEEE Transactions on Network and Service Management IEEE TNSM
- [6] **(Poster) BACKWARD: A Victim-Centric DDoS Detection and Mitigation Scheme in Programmable Data Plane** 📄
[Seoyul Oh](#), Sol Han, Hochan Lee, Sangheon Pack
IEEE Consumer Communications & Networking Conference CCNC 2023
- [7] **Holistic Orchestration for Edge-Native Applications: A Review** 📄
Taeyun Kim, [Seoyul Oh](#), Inho Cha, Seunghyun Lee, Haneul Ko, Sangheon Pack
IEEE International Conference on Information and Communication Technology Convergence ICTC 2022

Research Experience

Illinois Networked Systems and AI (INSAI) Lab, University of Illinois Urbana-Champaign IL, USA Jan 2025 - Current

Research Assistant

Project: First-Principles RL for Networking

- Designed a first-principles diagnostic framework to visualize and analyze RL training dynamics, identify key failure modes, and provide practical guidelines for reliably learning high-performance adaptive bitrate streaming policies [3]

Wonderful Wireless Lab, University of Illinois Urbana-Champaign IL, USA

Sep 2023 - Current

Research Assistant

Project: Decentralized Satellite Networking

- Proposed MP-LEO, a decentralized satellite network where participants share spare satellite capacity ensuring robust coverage without needing independent large constellations [1] [4]

Project: Energy-Aware Cellular Networking

- Developed EcoCell, a software-only middlebox solution that reduces base station energy consumption through traffic shaping techniques like subframe packing, UE segregation, and temporal traffic shifting [2]
- Validated EcoCell across applications, achieving energy savings with minimal impact on user experience

Mobile System, Security and Networking Lab, Purdue University West Lafayette, IN, USA

Aug 2022 - Feb 2023

Visiting Graduate Researcher (Advisor: Prof. Chunyi Peng)

Project: Better Mobility Support in 5G Cellular Networks

- Proposed a low-latency video delivery scheme for 5G cellular-connected drones
- Measured 5G performance in the sky through real-world experiments

Mobile Network and Communications Lab, Korea University Seoul, South Korea

Mar 2021 - Jul 2022

Research Assistant

Project: Network Security Utilizing Programmable Data Planes

- Led research on a P4-based victim centric DDoS detection and mitigation scheme [6]

Project: Development of an Open-source-based Edge Cloud Service Platform Prototype

- Analyzed future edge cloud service platform requirements
- Proposed a framework that performs holistic orchestration in edge cloud environments [5] [7]

Honors and Awards

Outstanding Teaching Assistant Award University of Illinois Urbana-Champaign Spring 2025

Korean Honor Scholarship (KHS) Ministry of Foreign Affairs, Korea (1,500 USD) Fall 2025

KASF-KIA Scholarship Korean American Scholarship Foundation (2,500 USD) Fall 2025

International R&D Program Grant for Graduate Students in Science & Technology Ministry of Science and ICT, Korea (12,580 USD) Jun 2022 - Feb 2023

Merit-based Scholarship School of Electrical Engineering, Korea University Spring 2020

Semester High Honors School of Electrical Engineering, Korea University 2020, 2019, 2017

Teaching Experience

CS538: Advanced Computer Networks University of Illinois, Urbana-Champaign Spring, Fall 2025

Teaching Assistant (Instructor: Francis Y. Yan)

EGRN322: Technology Trends for Establishing and Managing Business Korea University Fall 2021

Teaching Assistant (Instructor: Sangheon Pack)

KECE370: Digital Signal Processing Korea University Fall 2021

Korea University Center for Teaching and Learning (KUCTL) Tutor

Best Tutor Award & Best Team Award

References

- Prof. Francis Y. Yan Assistant Professor in Computer Science at UIUC [👤]
- Prof. Deepak Vasisht Assistant Professor in Computer Science at UIUC [👤]
- Prof. Chunyi Peng Associate Professor in Computer Science at Purdue University [👤]
- Prof. Sangheon Pack Professor in Electrical Engineering at Korea University [👤]